

MUHID HASSAN RISVY

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[GitHub](#) [◇ LinkedIn](#)

EDUCATION

New Jersey Institute of Technology

Fall 2024 - Current

Ph.D. (Ongoing) in Computer Science

Shahjalal University of Science & Technology

May 2023

B.Sc. in Computer Science and Engineering

CGPA: 3.56/4.00 (3.85 in the last two years).

RESEARCH EXPERIENCE

Security Analysis of Passkey

Fall 2025-

Currently analyzing how passkeys are saved in users' password managers. Analyzed credential storage security across multiple password managers and found significant metadata leakage, weak encryption technique which can reveal user's information like login data, URL visited, timestamps and other critical data. Moreover, developed a browser extension that shows compromised or at-risk passkeys for each relying party, i.e, website.

Browser Automation For Improving Security Guides

Summer 2025

Developed an AI-driven system to automatically validate and update security guides from organizations like EFF when platforms change their UI or settings. Built a hybrid local/remote browser automation system using open-source LLM models deployed on HPC clusters. Successfully integrated remote HPC GPU resources with local browser operations and used Playwright for secure web automation.

Detecting Typosquatting Attack on the Web

Fall 2024 - Spring 2025

The research domain focused on establishing the authenticity of software on the internet, with emphasis on typosquatting, a cybercrime where attackers register domains that are typo-versions of legitimate domains. I explored and analyzed defensive strategies and mitigation challenges, as well as the gaps in existing research.

Security Analysis of Software-Based Cryptocurrency Wallets: (Undergraduate Thesis): The research project aims to identify the security issues and privacy concerns in cryptocurrency wallets. It uses vulnerability assessment tools and user reviews to evaluate the perceived security and usability. Both manual and machine learning approaches are used to analyze user reviews and determine the wallets' effectiveness in keeping cryptocurrencies secure.

TEACHING EXPERIENCE

Teaching Assistant

Fall 2024 - Current

New Jersey Institute of Technology

CS 351 Introduction to Cybersecurity

Fall 2024

CS 388 Android Application Development

Spring 2025

CS 288 Intensive Programming in Linux

Summer 2025

CS 332 Principles of Operating Systems

Summer 2025

CS 288 Intensive Programming in Linux (two sections)

Fall 2025

PROJECTS

COVID-19 Vaccine Supply Chain | [GitHub](#)

A blockchain project implementing a private blockchain-powered supply chain to establish end-to-end traceability for the COVID-19 vaccine, while guaranteeing secure distribution and transportation.

Technology: Hyperledger Fabric, Node.js, JavaScript, Apache CouchDB.

Passkey Health Manager | [GitHub](#)

A Chrome extension that manages passkeys with advanced health auditing features to track passkey security, detect weak keys using hash collision detection, monitor share counts, and maintain comprehensive export history.

Technology: React, TypeScript, Chrome Extension API.

E-commerce & Bank | [GitHub](#)

API-based website to simulate the functionalities of e-commerce services among 3 different organizations i.e., Customer, Supplier, and a Bank for instant transactions.

Technology: Node.js, Express.js, MongoDB.

RELATED COURSES AND SKILLS

Courses	CS 698: Human Factors in Security and Privacy (A), CS 756: Mobile Computing and Sensor Networks (A), CS 645: Security and Privacy in Computer Systems (A)
Programming Languages	Solved 850+ problems on CodeForces , LeetCode , Vjudge , etc. C, C++, Java, Python, JavaScript, MATLAB

ACCOMPLISHMENTS

Bangladesh Blockchain Olympiad 2022 <i>Finalist</i>	June 2022
Bangladesh Blockchain Olympiad 2021 <i>Finalist</i>	February 2021

EXTRA-CURRICULAR ACTIVITIES

Soccer: Champion, Intra IICT Tournament, 2019

Table Tennis: Runners-Up, Intra-Department Tournament, 2022